

Al Musafir International

Umrah Travel ERP — Complete Project Map

This document is a complete reference guide for the Al Musafir International Umrah travel ERP. It maps every module, every user role, and how data flows between the staff dashboard, public website, and API server. Use it to understand the system at a glance, onboard new developers, or review the full development cost estimate.

WHAT'S INSIDE

1. System Architecture Overview — how the 3 apps talk to each other
2. User Roles & Permissions — who can do what
3. ERP Module Directory — every screen, grouped by function
4. Customer Journey Map — from visitor to download
5. API Reference Map — which endpoint powers which screen
6. Database Table Directory — what data is stored and where
7. Key Operational Workflows — step-by-step process guides
8. Developer Quick-Reference — where to make changes in the code
9. Estimated Development Cost — hours & cost per module in USD & PKR

BUILT WITH:

Node.js 24

TypeScript

React + Vite

Express 5

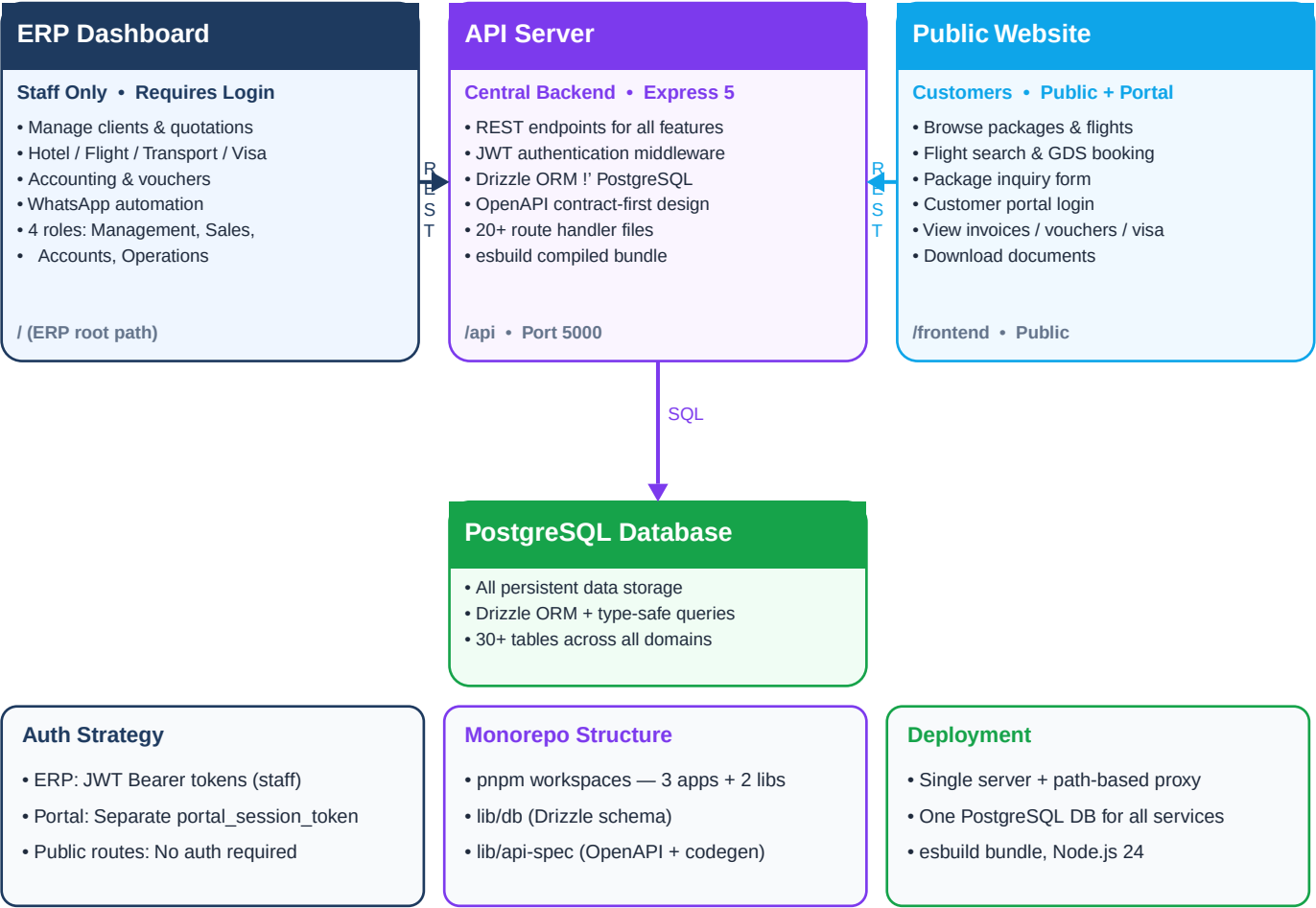
PostgreSQL

Drizzle ORM

pnpm Workspaces

System Architecture

How the three applications connect to each other and to the database



User Roles & Permissions

Who uses the system and what each role can access

Role	Who	What They Can Access
Management	Owner / Senior Management	Full access to everything — all modules, all reports, all settings, can create/edit/delete any record
Sales	Sales Executives	Clients, quotations, hotel requests, follow-ups. Cannot access accounting vouchers or system settings
Accounts	Finance / Accounts Team	Full accounting access: vouchers, ledger, P&L, balance sheet, invoices. Cannot change system settings
Operations	Operations Staff	Hotels, flights, transport, visa, bookings. View-only accounting. Cannot change system settings
Portal Customer	End Customers (self-registered)	Customer portal only: own bookings, invoices, vouchers, visa docs, transport, downloads. No ERP access
Portal DC / Agent	Distributor / Sub-agent	Portal DC login: own client bookings and documents. Separate from end-customer portal. No ERP access

HOW ROLES ARE ASSIGNED

ERP staff roles are set by a Management user in the Users section of the ERP (/users). A new staff member receives a temporary password and must change it on first login. Portal customers self-register at /portal-register on the public website — their account starts as 'Pending Approval' and a staff member must activate it from /portal-users in the ERP before they can log in.

SECURITY FEATURES

Forced Password Change	New staff accounts are flagged 'must change password'. The server blocks all API calls (except /change-password) until the password is changed.
Bcrypt Password Hashing	All passwords are stored as bcrypt hashes (\$2b\$). Plain-text passwords from old accounts are automatically re-hashed on server startup.
Login Audit Log	Every login attempt (success or failure), password change, and reset is written to the auth_audit_log table with timestamp and IP.
Separate Portal Auth	Portal customers use a completely separate authentication system (portal_session_token) — a portal token cannot access ERP endpoints.
Route-Level Guards	Every ERP API endpoint requires requireAuth middleware. Public routes (packages, currency rates, website config) are explicitly opted-out.

ERP Module Directory

Every screen in the staff dashboard, grouped by function

CRM & Sales

Dashboard	/	Summary stats: today's invoices, pending quotations, alerts
Clients	/crm	Master list of all clients/parties — add, edit, view profile
Client Profile	/crm/:id	Full client detail: bookings, invoices, documents, history
Follow-Ups	/crm/follow-ups	Pending follow-up reminders for the sales team
Quotations	/quotations	Create & manage price quotations; PDF generation
Pending Quotes	/quotations/pending	Quotations awaiting client confirmation
Hotel Requests	/hotel-requests	Incoming hotel availability requests from clients

Operations

Hotels Directory	/hotels	Hotel master list: name, city, stars, rooms, WhatsApp
Vendors	/vendors	Vendor/supplier master list
Transport	/transport	Transport bookings: vehicle type, vendor, cost, status
Flights	/flights	Flight records: sector, airline, PNR, pax, fares
Flight Cancellations	/flights/cancellations	Cancelled flight tracker
BSP Report	/flights/bsp-report	IATA BSP billing summary report
Passenger List	/flights/passengers	All passenger records across flights
Visa Applications	/visa	Visa tracking: type, applicant, status, documents

Finance & Accounting

Invoice List	/accounting/invoices	All hotel DN invoices — list, filter, search
Hotel Invoice Form	/accounting/hotel-invoice/:id	Create/edit hotel invoice; 3 print formats
Vouchers	/accounting/vouchers	Double-entry vouchers: RV/PV/JV/CV types
General Ledger	/accounting/ledger	Account-level transaction ledger
Trial Balance	/accounting/trial-balance	Debit/credit balances for all accounts
P&L Statement	/accounting/pnl	Profit & Loss report by date range
Balance Sheet	/accounting/balance-sheet	Assets, liabilities, equity snapshot
Financial Years	/financial-years	Open/close accounting periods
Currency Settings	/currency-settings	Exchange rates, multi-currency config

WhatsApp & Communications

WhatsApp Inbox	/whatsapp-inbox	Read incoming WhatsApp messages
Bot Campaigns	/bot-campaign	Automated message campaigns to groups/contacts
Media Library	/media-library	Uploaded images/docs for WhatsApp sends

Administration & Settings

Users	/users	ERP staff accounts: add, role, reset password
Portal Users	/portal-users	Customer portal accounts: approve, view, manage
Website Settings	/website-settings	Branding, WhatsApp buttons, homepage config
ERP Settings	/erp-settings	ERP-level configuration
Automation Settings	/automation-settings	WhatsApp automation rules
GDS Settings	/gds-settings	Flight GDS API credentials
AI Settings	/ai-settings	OpenAI API key for OCR/AI features

Customer Journey Map

The end-to-end path a customer takes from first visit to document download



API !” Module Map

Which API route file powers which part of the system

API Route File	ERP / Website Page	What it Does
auth.ts	ERP Login, /users (change/reset password)	JWT login, logout, me, change-password, reset-password
users.ts	ERP /users	Staff user CRUD, role management
portal.ts + portal-customer.ts	Portal /portal-login, /portal/dashboard	Portal auth, dashboard, bookings, invoices, docs
portal-users.ts	ERP /portal-users	Approve/reject/list portal accounts
clients.ts	ERP /crm, /crm/:id	Client/party CRUD, search, profile data
quotations.ts	ERP /quotations, /quotations/pending	Quotation create/edit/status/PDF
booking-inquiries.ts	ERP, Public website inquiry form	Store & list flight/booking inquiries
package-inquiries.ts	Public website package inquiry	Store & list package inquiry submissions
hotel-invoices.ts	ERP /accounting/invoices, /hotel-invoice/:id	DN invoice CRUD, next-DN generation
hotels.ts	ERP /hotels, Public /public/hotels	Hotel master CRUD, public listing
vendors.ts	ERP /vendors	Vendor/supplier CRUD
flights.ts	ERP /flights and sub-pages	Flight record CRUD, BSP, cancellations, passengers
transport.ts	ERP /transport	Transport booking CRUD
visa.ts	ERP /visa	Visa application tracking CRUD
vouchers.ts	ERP /accounting/vouchers	Double-entry voucher create/edit/post
accounting.ts	ERP /accounting/ledger, trial-balance	Chart of accounts, ledger, trial balance
accounting-reports.ts	ERP /accounting/pnl, /balance-sheet	P&L and balance sheet reports
financial-year.ts	ERP /financial-years	Financial year CRUD, open/close
currency-settings.ts	ERP /currency-settings, /api/currency/rates	Exchange rates, live rate proxy (public)
whatsapp.ts	ERP /whatsapp-inbox, /bot-campaign	WhatsApp send, session, groups, inbox
automations.ts + ai-settings.ts	ERP /automation-settings, /ai-settings	Automation rules, OpenAI key management
website-config.ts	ERP /website-settings, /api/website-config (public)	Branding, homepage config (public read)

Database Table Directory

Every table in PostgreSQL — what it stores and how it connects

Core / Authentication

users	ERP staff accounts	Linked to: vouchers, invoices, quotations (salesman)
portal_users	Customer/DC portal accounts	Linked to: clients (party)
auth_audit_log	Every login/logout/password event	Linked to: users

CRM & Sales

clients	Client/party master	Linked to: quotations, invoices, portal_users
quotations	Price quotations with status	Linked to: clients, users (salesman)
hotel_requests	Hotel availability requests	Linked to: clients, users
booking_inquiries	Inbound flight/booking inquiries	Standalone + client link
package_inquiries	Inbound package inquiries from website	Standalone

Operations

hotels	Hotel master: name, city, stars, rooms	Linked to: hotel_invoices
vendors	Supplier/vendor master	Linked to: hotel_invoices, transport_bookings
hotel_invoices	Hotel DN invoices	Linked to: clients, vendors, hotels, users
flights	Flight segment records	Linked to: clients, passengers
group_tickets	Group flight ticket records	Linked to: flights
transport_bookings	Transport bookings	Linked to: clients, vendors
visa_applications	Visa applications	Linked to: clients
passenger_documents	Uploaded passport/visa scans	Linked to: clients, portal_users

Finance & Accounting

accounts	Chart of accounts (assets/liab/equity/income/expenses)	Linked to: voucher_lines
vouchers	ERP/PPV/CV voucher headers	Linked to: accounts, users
voucher_lines	Individual debit/credit lines	Linked to: vouchers, accounts
financial_years	Accounting period definitions	Linked to: vouchers
currency_settings	Exchange rates and base currency config	Standalone

Communications & System

whatsapp_groups	WhatsApp group targets	Standalone
whatsapp_messages	Sent/received message log	Linked to: clients
bot_campaigns	Automated campaign definitions	Linked to: whatsapp_groups
automation_configs	Automation rule settings	Standalone
media_library	Uploaded media files for WhatsApp	Standalone
website_config	Key-value store for branding & settings	Standalone

Key Operational Workflows

Step-by-step process guides for the most common staff tasks

1. Create & Confirm a Quotation

- 1 Go to CRM (/crm) and open the client's profile, or create a new client.
- 2 Click 'New Quotation' !' fill in the package details, hotel, nights, pax count.
- 3 Add hotel cost (payable) and selling price (receivable) — system calculates profit.
- 4 Save as 'Draft'. Review and update if needed.
- 5 Change status to 'Sent' when shared with the client.
- 6 When client confirms, update status to 'Confirmed'. This triggers an option date alert.
- 7 Proceed to raise a Hotel Invoice (DN) from the Accounting module.

2. Raise a Hotel Invoice (DN)

- 1 Go to Accounting !' Invoices (/accounting/invoices) !' click 'New Hotel Invoice'.
- 2 Select the client (Party) and hotel vendor from the dropdowns.
- 3 Fill in hotel details: check-in/out dates (Nights auto-calculates), rooms, bed type.
- 4 In the Calculation section, enter the Rate per Night per Room (SAR).
- 5 System auto-multiplies: Rate × Nights × Rooms = Receivable Total.
- 6 Enter the PKR exchange rate or click 'live' to fetch the current rate.
- 7 Click Accept. The DN number (e.g. DN-0042) is assigned automatically.
- 8 Print the formal invoice, DN invoice, or accommodation voucher using the print buttons.

3. Post a Payment Voucher (RV / PV)

- 1 Go to Accounting !' Vouchers (/accounting/vouchers) !' click 'New Voucher'.
- 2 Select type: RV (Receipt Voucher = money received), PV (Payment Voucher = money paid).
- 3 Enter Date, Narration (description of the transaction).
- 4 Add at least 2 lines: one DEBIT line and one CREDIT line. Totals must balance (DR = CR).
- 5 Select the account for each line from the Chart of Accounts dropdown.
- 6 Click Save. The voucher is posted to the General Ledger immediately.
- 7 Verify in Ledger (/accounting/ledger) by filtering on the relevant account.

4. Send a WhatsApp Message to a Vendor

- 1 Go to WhatsApp Inbox (/whatsapp-inbox).
- 2 Ensure WhatsApp is connected (scan QR code once; session persists on server).
- 3 Click 'New Message' !' select the vendor's saved contact/group.
- 4 Type the message or select a template from Media Library.
- 5 Click Send. Message appears in the inbox log with delivery status.

5. Portal Customer: Register !' Access Documents

- 1 Customer visits the public website and clicks 'Customer Portal' or /portal-register.
- 2 Fills in name, email/phone, and password. Account created with status 'Pending Approval'.
- 3 Staff goes to ERP !' Portal Users (/portal-users) and clicks 'Approve'.
- 4 Customer receives confirmation and logs in at /portal-login.
- 5 Portal dashboard shows their bookings, invoices, vouchers, visa status, transport.
- 6 Customer downloads travel documents from /portal/downloads.

Developer Quick-Reference

Where to find or change anything in the codebase

If You Need To...	Look In / Go To	Notes
Add a new ERP page / screen	artifacts/umrah-erp/src/pages/	Create a new .tsx file, add a route in App.tsx
Add a new public website page	artifacts/frontend/src/pages/	Create .tsx file, add route in frontend App.tsx
Add or change a database table	lib/db/src/schema/	Edit the Drizzle schema file, then run: pnpm --filter @workspace/db run push
Add a new API endpoint	artifacts/api-server/src/routes/	Add route handler, register in index.ts, update OpenAPI spec
Update the OpenAPI spec	artifacts/api-spec/openapi.yaml	Edit spec, then run: pnpm --filter @workspace/api-spec run codegen
Change company branding (print)	ERP ! Website Settings (/website-settings)	Update keys in website_config table via the settings UI
Change company logo	ERP ! Website Settings ! Logo Upload	Uploads to object storage; keys: logo_url, print_logo_url
Add a WhatsApp automation rule	ERP ! Automation Settings (/automation-settings)	Configure trigger/action pairs in the UI
Change exchange rates	ERP ! Currency Settings (/currency-settings)	Manual entry or live fetch from currency API
Reset a staff password	ERP ! Users (/users) ! Reset Password	Sends a temporary password; user must change on login
Approve a portal customer	ERP ! Portal Users (/portal-users)	Click Approve on pending registration
View accounting ledger	ERP ! Accounting ! Ledger (/accounting/ledger)	Filter by account code or date range
Generate a P&L report	ERP ! Accounting ! P&L (/accounting/pnl)	Select date range; exports to print
Check server logs	Workflow console / deployment logs	Use the Replit workflow console for live logs
Run a DB migration (dev)	Terminal: pnpm --filter @workspace/db run push	Pushes schema changes to the development database
Build the API server	Terminal: pnpm --filter @workspace/api-server run build	Compiles TypeScript via esbuild to dist/
Run full typecheck	Terminal: pnpm run typecheck	Typechecks all workspace packages
Regenerate API hooks & schemas	Terminal: pnpm --filter @workspace/api-spec run codegen	Regenerates React Query hooks from OpenAPI spec

PROJECT FOLDER STRUCTURE

artifacts/api-server/ Express API server — all backend code	artifacts/umrah-erp/ Staff ERP dashboard (React + Vite)
artifacts/frontend/ Public website + customer portal (React + Vite)	lib/db/ Drizzle ORM schema + migrations (shared library)
lib/api-spec/ OpenAPI contract + generated client hooks	scripts/ Utility scripts (e.g. this PDF generator)

Estimated Development Cost

Agent-rate estimate (all 20 modules) • \$5/hr internal rate • 1 USD = 290 PKR

Module / Area	Features Included	Hrs	Status	Cost (USD)	Cost (PKR)
Project Setup & Infrastructure	Monorepo, API server, DB schema, CI, deployment config	40	● ' Done	\$200	Rs 58,000
Authentication & User Roles	Login, sessions, roles, forced password change, audit log	30	● ' Done	\$150	Rs 43,500
CRM / Clients	Client list, profile, follow-ups, hotel requests	35	● ' Done	\$175	Rs 50,750
Quotations	Create/edit quotation, PDF generation, status workflow, pending approvals	45	● ' Done	\$225	Rs 65,250
Hotel Invoices (DN)	Invoice form, rate×nights auto-calc, 3 print formats, status tracking	50	● ' Done	\$250	Rs 72,500
Flight Management	Flight records, BSP report, cancellations, passenger manifest	40	● ' Done	\$200	Rs 58,000
Transport Bookings	Transport invoice form, vendor assignment, status tracking	25	● ' Done	\$125	Rs 36,250
Visa Applications	Visa tracking, document upload, status workflow	25	● ' Done	\$125	Rs 36,250
Hotels & Vendors Directory	Hotel master list, vendor list, room types, WhatsApp numbers	20	● ' Done	\$100	Rs 29,000
Double-Entry Accounting	Chart of accounts, RV/PV/JV/CV vouchers, ledger, trial balance, P&L, balance sheet	70	● ' Done	\$350	Rs 101,500
Financial Years	Year open/close, period locking, reporting periods	15	● ' Done	\$75	Rs 21,750
Currency Settings	Multi-currency support, live rate fetching, PKR conversion	15	● ' Done	\$75	Rs 21,750
WhatsApp Automation	Inbox, bot campaigns, media library, group messaging, session persistence	60	● ' Done	\$300	Rs 87,000
Customer Portal	Portal login/register, dashboard, invoices, vouchers, visa, transport, downloads	55	● ' Done	\$275	Rs 79,750
Public Website	Landing page, package listings, flight search, GDS booking, package inquiry	50	● ' Done	\$250	Rs 72,500
Print & Branding System	Company branding config, print layouts for all document types, logo upload	20	● ' Done	\$100	Rs 29,000
AI / OCR Integration	Document scanning stub, AI settings page, OpenAI key management	15	● ' Done	\$75	Rs 21,750
ERP Settings & Admin	Website settings, automation, GDS/AI/ERP configurations	20	● ' Done	\$100	Rs 29,000
Portal Users Management	Staff approval workflow, portal user list, status management	15	● ' Done	\$75	Rs 21,750
Testing, Bug Fixes & Polish	Cross-module testing, UI polish, error handling, security hardening	50	● % Pending	\$250	Rs 72,500

● AGENT-WORKED (done + active):	645 hrs • \$3,225 USD • Rs 935,250 PKR
● REMAINING ESTIMATE (pending):	50 hrs • \$250 USD • Rs 72,500 PKR
GRAND TOTAL (695 hrs @ \$5/hr):	USD 3,475 PKR 1,007,750

NOTE: Costs calculated at the agent rate of \$5/hr. Market rate for an equivalent mid-senior developer team would be \$35–50/hr. Does not include third-party service costs (hosting, WhatsApp Business API, SMS, GDS/Amadeus fees, domain/SSL). Testing & Polish module is ongoing; final hours may vary.